

Day 1: Math Foundations

- **Math Foundations** — Linear algebra through differential equations — MML Part I
- 3 hours lecture + practical
- Slides, notes, and code on the course site

- The Mathematical Foundation
- Linear Algebra
- Analytic Geometry
- Vector Calculus & Decompositions
- Integration & Differentiation
- Probability & Distributions
- Differential Equations

The Mathematical Foundation



Four Pillars of ML

- Data as vectors; tables of data as matrices (linear algebra)
- Similarity via norms and inner products (analytic geometry)
- Structure via matrix decompositions; optimization via gradients (Ch. 4–5)
- Uncertainty via probability; dynamics via differential equations
- Source: Deisenroth, Faisal & Ong — [Mathematics for Machine Learning](#)

Four Pillars of ML — illustration

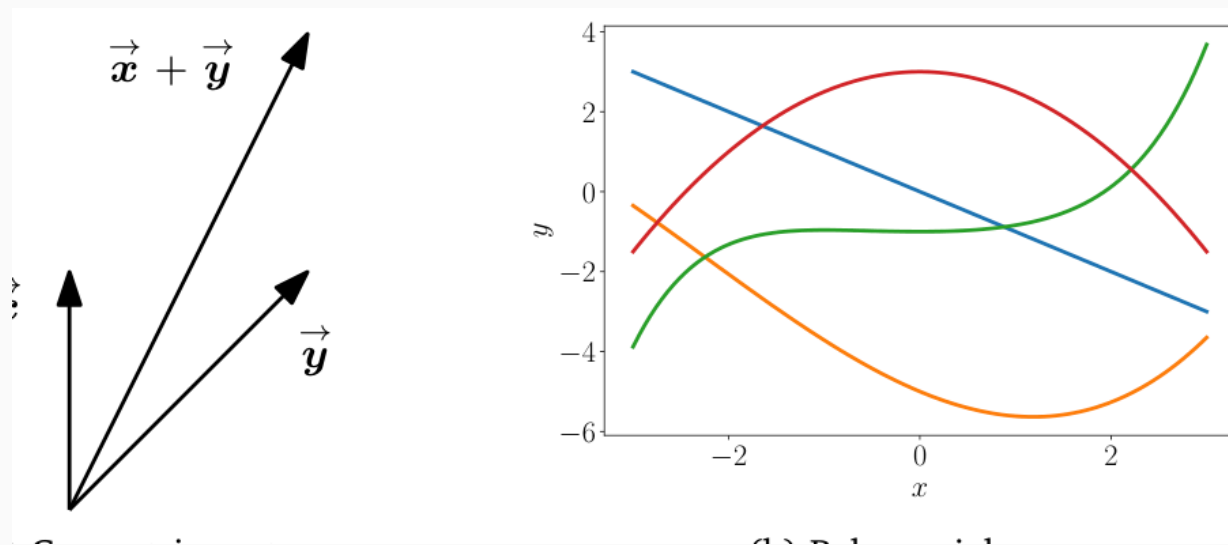


Figure 2.1
Different types of vectors. Vectors can be surprising objects, including (a) geometric vectors and (b) polynomials.

The Mathematical Foundation — Four Pillars of ML (source: course materials)

Roadmap for Today

- 1. Linear algebra — equations, matrices, vector spaces
- 2. Analytic geometry — norms, inner products, projections
- 3. Vector calculus & matrix decompositions
- 4. Integration & differentiation (**There and Back Again**)
- 5. Probability & distributions
- 6. ODE/SDE crash course (diffusion book appendix)

Linear Algebra



From Linear Equations to Matrices

- System $Ax = b$: each row is one linear constraint on unknowns $x \in \mathbb{R}^n$
- Coefficient matrix $A \in \mathbb{R}^{m \times n}$ encodes all equations at once
- Solution sets are affine subspaces (when consistent)
- Matrix-vector product: each output component is an inner product of a row with x

From Linear Equations to Matrices – illustration

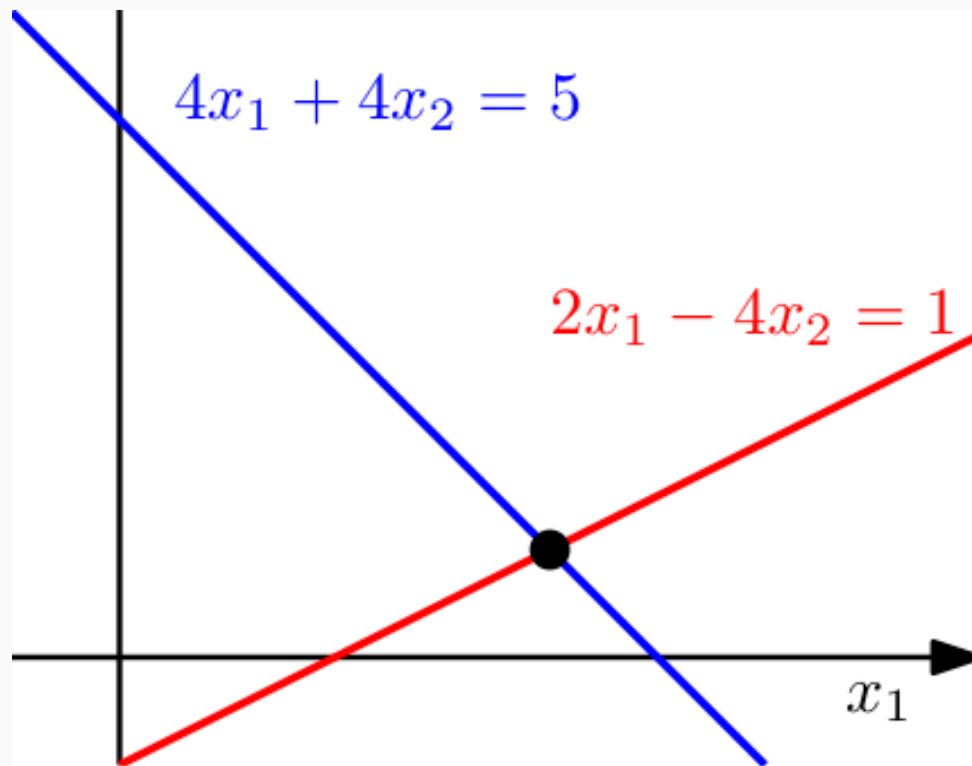
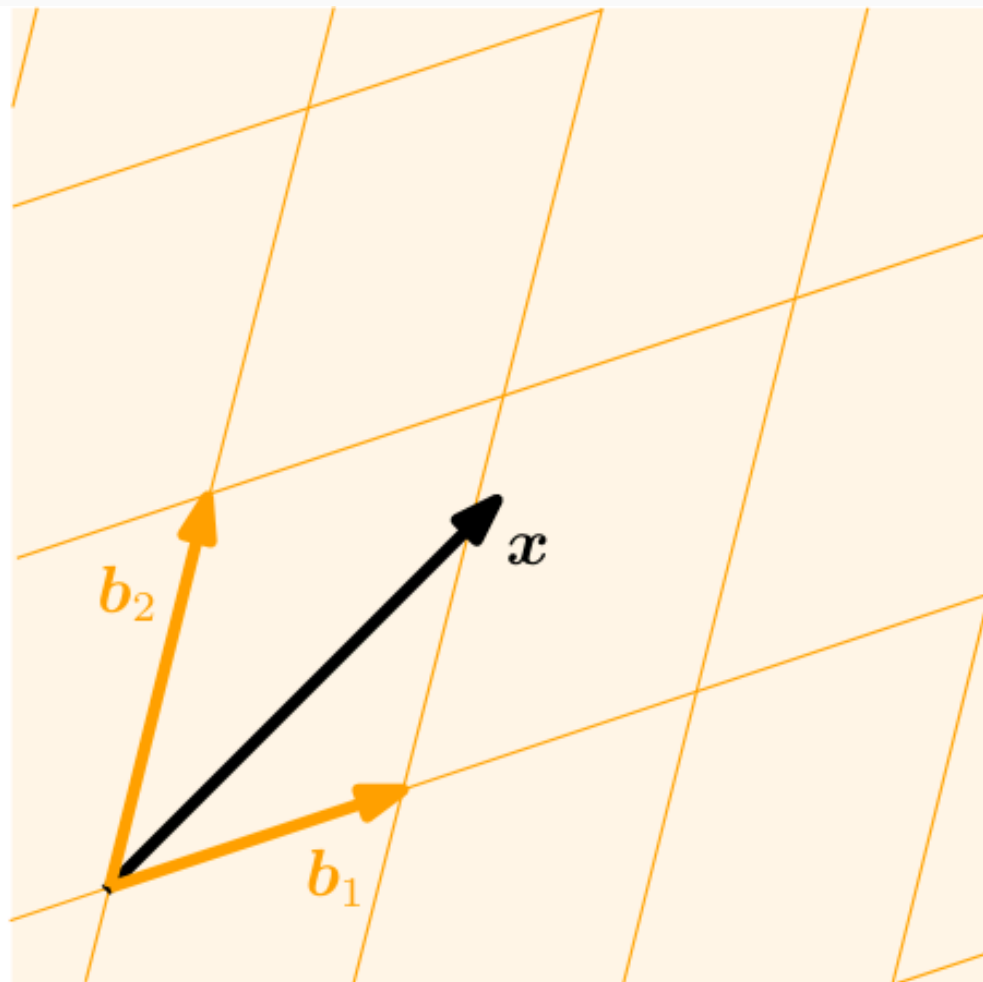


Figure 2.3 The solution space of a system of two linear equations with two variables can be geometrically interpreted as the intersection of two lines. Every linear equation represents a line.

What Is a Matrix?

- A matrix is a linear map $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ written as $y = Ax$
- Columns of A show where basis vectors land — geometric action of the map
- Composition of maps = matrix multiplication: $(AB)x = A(Bx)$
- Transpose swaps rows/columns; inverse (when it exists) undoes the map

Figure 2.8 Two different coordinate systems defined by two sets of basis vectors. A vector x has different coordinate representations depending on which coordinate system is chosen.



What Is a Matrix? — illustration

Linear Algebra — What Is a Matrix? (source: course materials)

- Vector space: closed under addition and scalar multiplication
- Linear combination: $\sum_i \alpha_i v_i$; span = all reachable combinations
- Linear independence: no vector is a combination of the others
- Basis: minimal spanning set; dimension = number of basis vectors
- Rank of A = dimension of column space = dim of row space

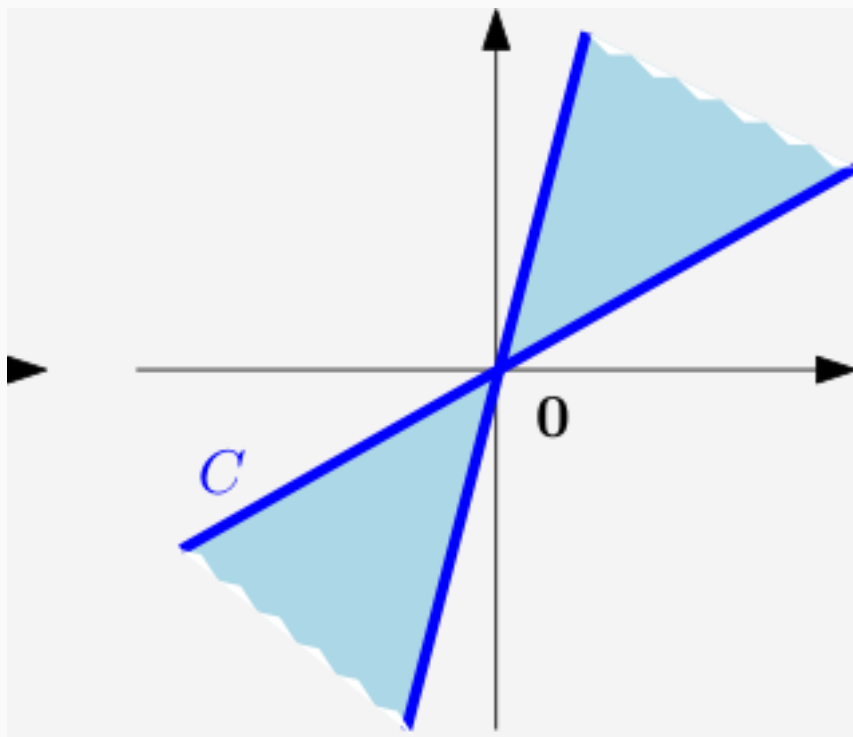


Figure 2.6 Not all subsets of $\mathbb{R}^2$ are subspaces. In A and C , the closure property is violated; B does not contain 0 . Only D is a subspace.

Solving & Interpreting $Ax = b$

- Gaussian elimination / LU factorization (algorithmic)
- Column picture: b must lie in span of columns of A
- Row picture: intersection of hyperplanes
- Null space: all x with $Ax = 0$ — directions the map collapses

Solving & Interpreting $Ax = b$ – illustration

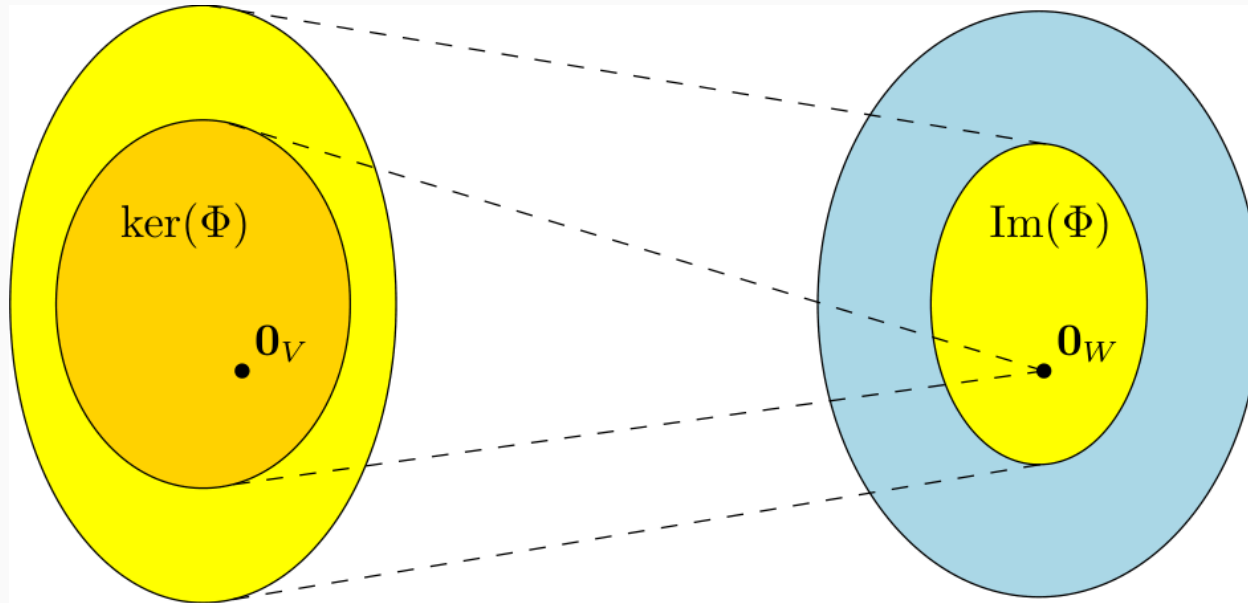


Figure 2.12 Kernel and image of a linear mapping $\Phi : V \rightarrow W$.

Linear Algebra – Solving & Interpreting $Ax = b$ (source: course materials)

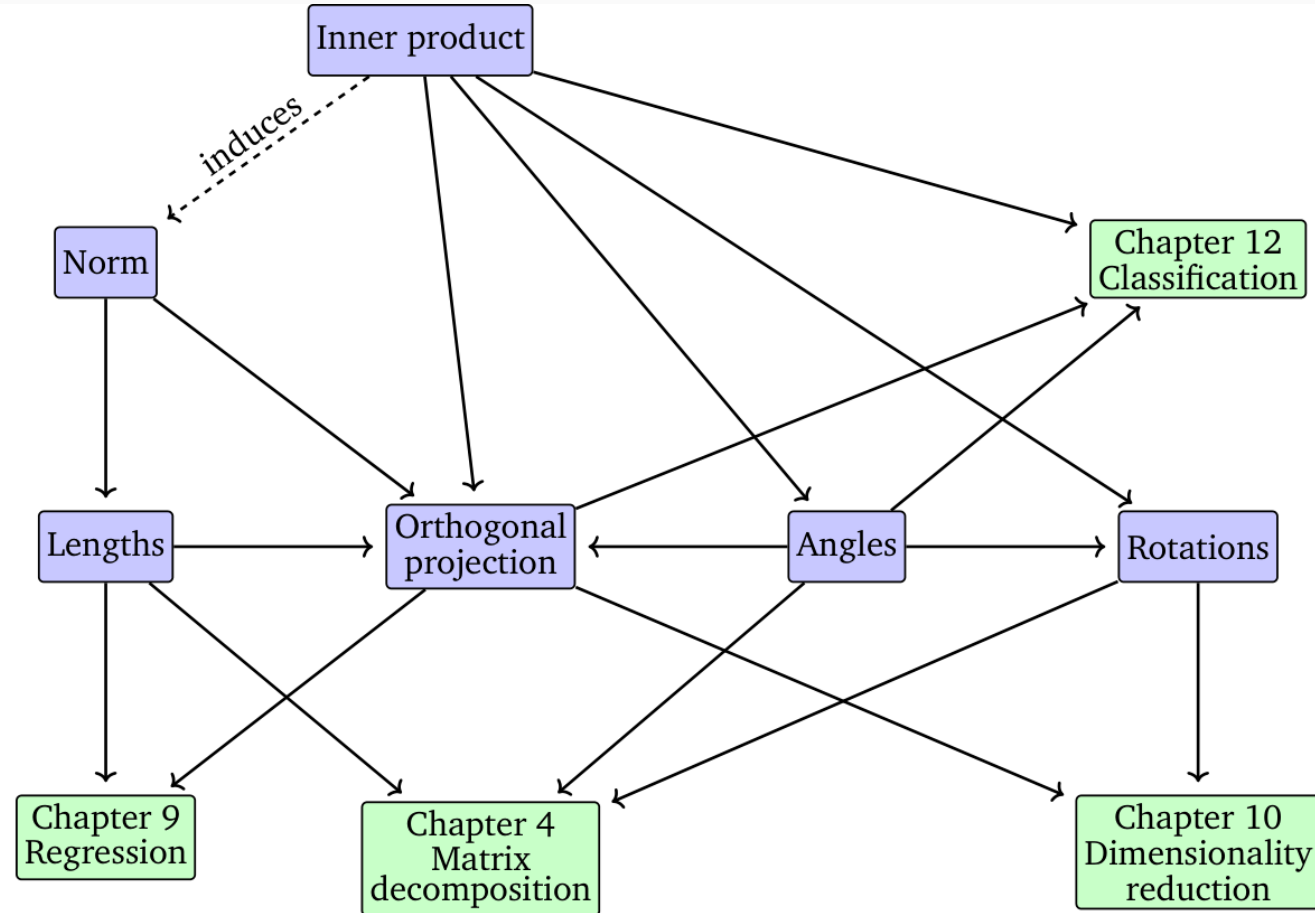
Analytic Geometry

Why Similarity Matters

- ML predictors should assign similar outputs to similar inputs
- Vectors encode objects; we need a numeric notion of ‘close’ or ‘aligned’
- Norms measure length/magnitude; inner products measure alignment
- Builds directly on linear-algebra vector operations

Why Similarity Matters – illustration

Figure 3.1 A mind map of the concepts introduced in this chapter, along with when they are used in other parts of the book.



- L_2 (Euclidean): $\|x\|_2 = \sqrt{\sum_i x_i^2}$ – rotation-invariant length
- L_1 : $\|x\|_1 = \sum_i |x_i|$ – sparsity-inducing geometry
- L_∞ : $\|x\|_\infty = \max_i |x_i|$
- Unit ball shapes differ – choice of norm changes regularization (Day 2)

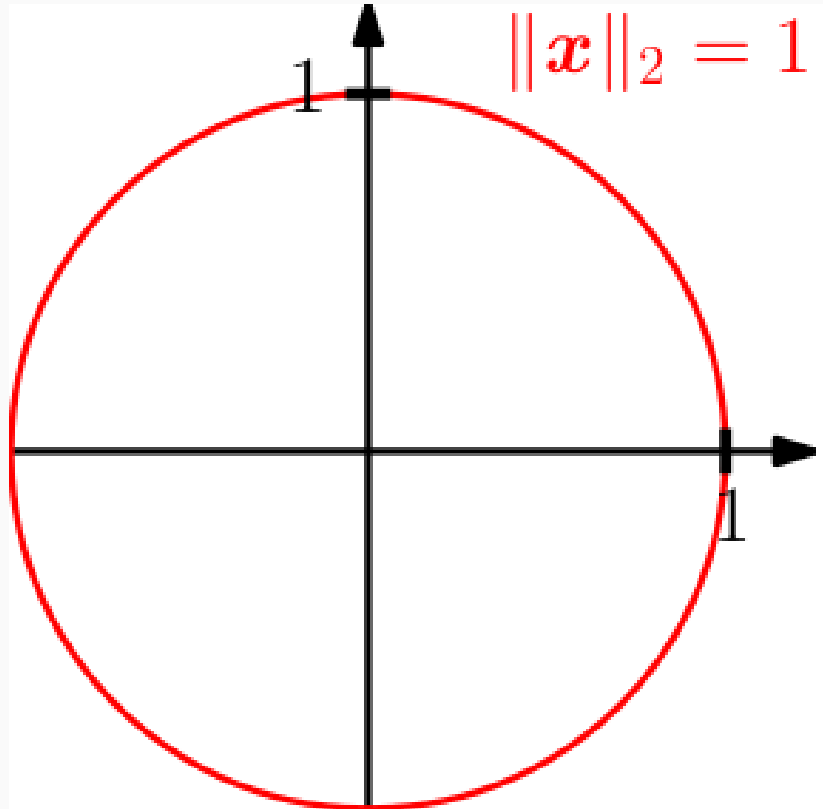


Figure 3.2 Triangle inequality.

Inner Product & Angles

- Inner product $\langle x, y \rangle = x^T y$ (standard dot product)
- Cauchy-Schwarz: $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$
- Angle: $\cos \theta = \frac{\langle x, y \rangle}{\|x\|_2 \|y\|_2}$
- Orthogonal vectors: $\langle x, y \rangle = 0$

Inner Product & Angles — illustration

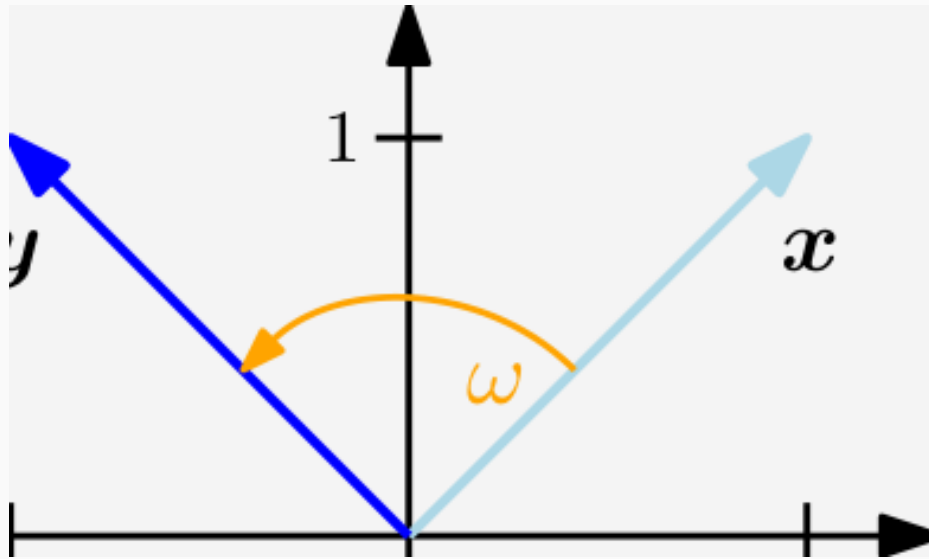


Figure 3.6 The angle ω between two vectors x, y can change depending on the inner product.

Analytic Geometry — Inner Product & Angles (source: course materials)

- Orthogonal projection of y onto line spanned by x : $\hat{y} = (x^T \frac{y}{x^T x})x$
- Projection matrix onto column space of A : $P = A(A^T A)^{-1} A^T$
- Least squares: minimize $\|Aw - y\|_2^2$ – project y onto span(columns of A)
- Normal equations: $A^T Aw = A^T y$

Projections – illustration

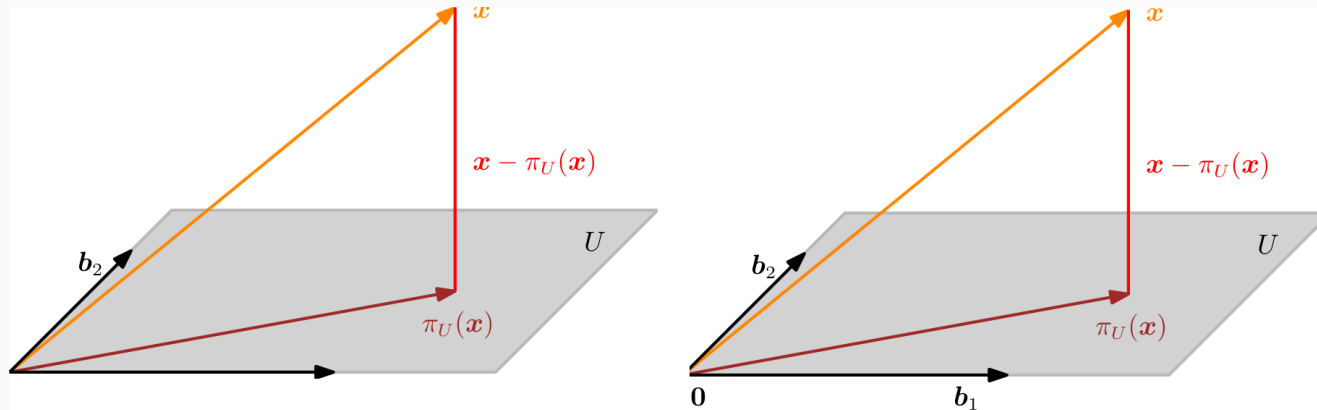


Figure 3.11
Projection onto a two-dimensional subspace U with basis b_1, b_2 . The projection $\pi_U(\mathbf{x})$ of $\mathbf{x} \in \mathbb{R}^3$ onto U can be expressed as a linear combination of b_1, b_2 and the displacement vector $\mathbf{x} - \pi_U(\mathbf{x})$ is orthogonal to both b_1 and b_2 .

Analytic Geometry – Projections (source: course materials)

Vector Calculus & Decompositions

- Scalar $f(x)$: gradient $\nabla f(x) \in \mathbb{R}^n$ — direction of steepest ascent
- Vector $f(x) \in \mathbb{R}^m$: Jacobian $J \in \mathbb{R}^{m \times n}$, $J_{ij} = \partial \frac{f_i}{\partial} x_j$
- Matrix-valued functions: treat as stacked scalar functions
- Chain rule: $\nabla_x (g \circ f) = J_f^T \nabla_g$ (reverse-mode building block)

Gradients & Jacobians — illustration

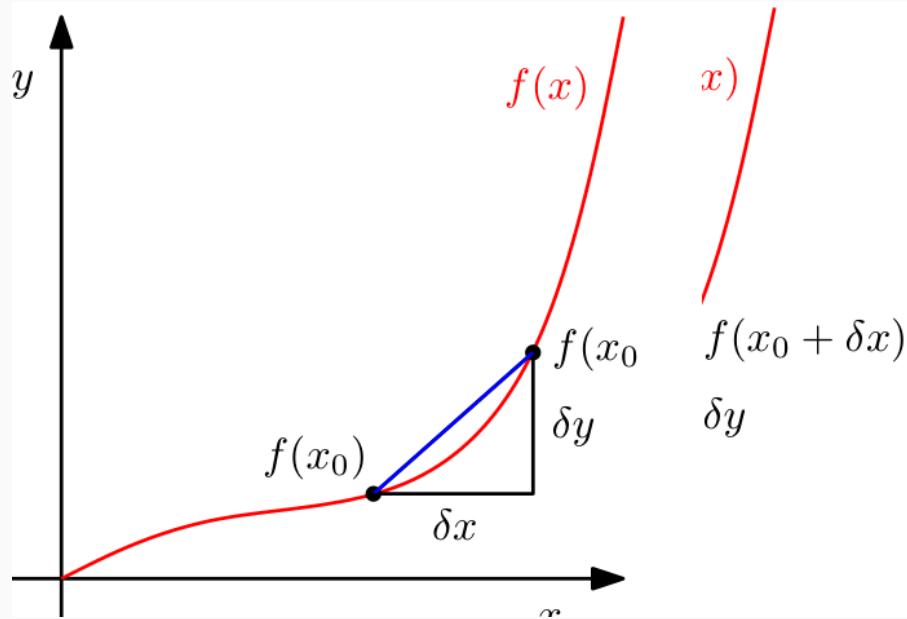


Figure 5.3 The average incline of a function f between x_0 and $x_0 + \delta x$ is the incline of the secant (blue) through $f(x_0)$ and $f(x_0 + \delta x)$ and given by $\delta y / \delta x$.

Vector Calculus & Decompositions — Gradients & Jacobians (source: course materials)

- $\nabla_x (a^T x) = a$; $\nabla_x (x^T A x) = (A + A^T)x$
- $\nabla_A \|Ax - y\|_2^2 = 2(Ax - y)x^T$
- Hessian $H = \nabla^2 f$ — second-order curvature for Newton methods
- These identities appear constantly in backprop and closed-form solutions

Useful Matrix Derivatives — illustration

original function

$$f(x) =$$

$$\sin(x) + \cos(x)$$

(black, solid) is

approximated by

Taylor polynomials

(dashed) around

$$x_0 = 0.$$

Higher-order Taylor

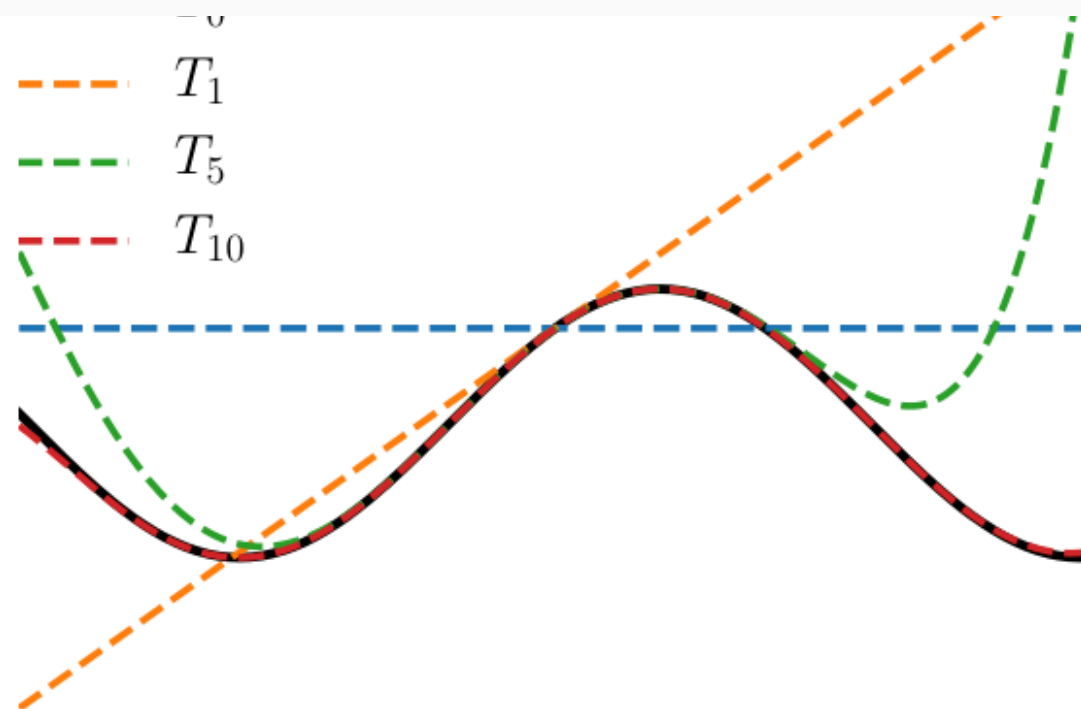
polynomials

approximate the

function f better

and more globally.

T_{10} is already



Useful Matrix Derivatives — illustration

Vector Calculus & Decompositions — Useful Matrix Derivatives (source: course materials)

Eigendecomposition

- $Av = \lambda v$ – eigenvector direction preserved, scaled by λ
- Symmetric A : real eigenvalues, orthogonal eigenvectors
- Spectral theorem: $A = Q\Lambda Q^T$ – principal axes of quadratic forms
- Condition number from eigenvalue spread – numerical stability

Eigendecomposition — illustration

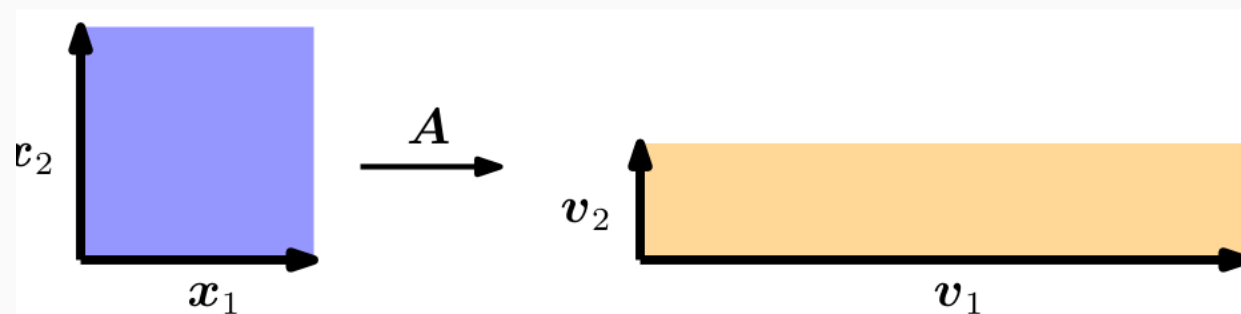
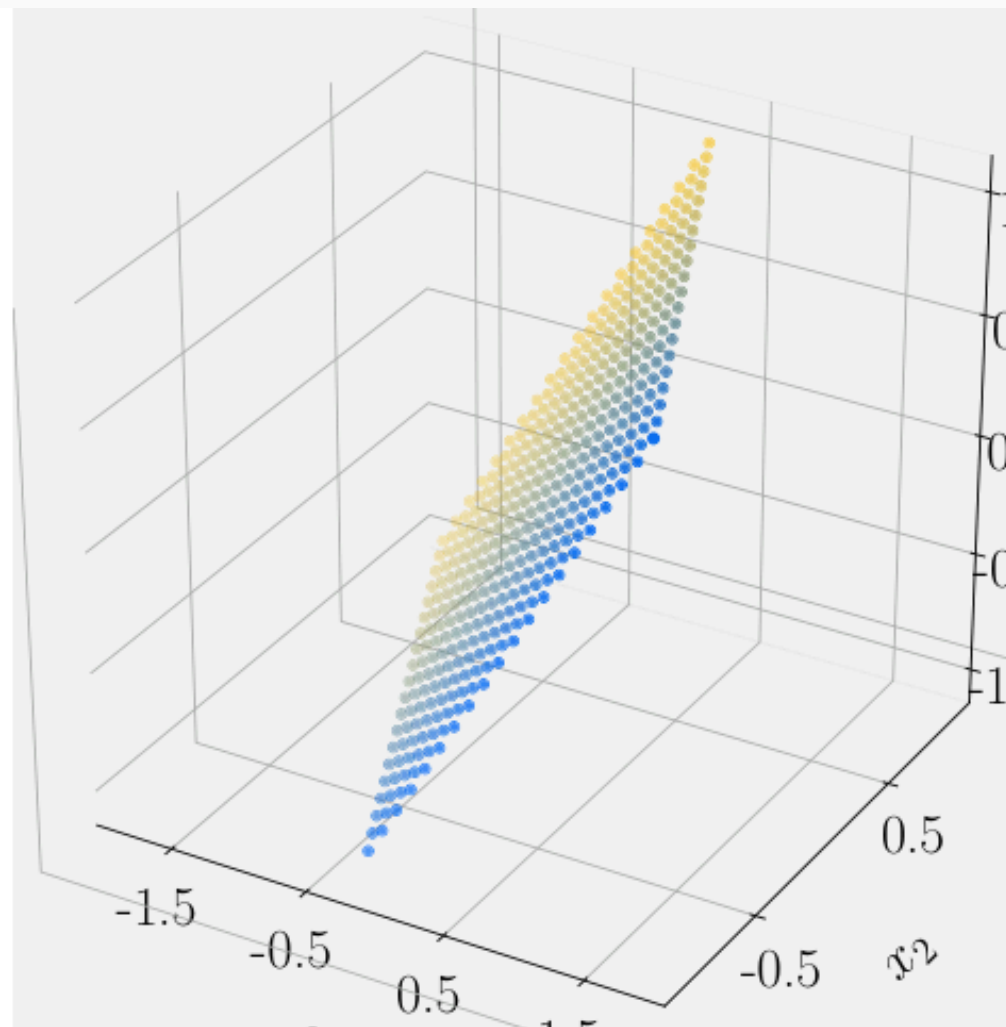


Figure 4.6
Geometric interpretation of eigenvalues. The eigenvectors of A get stretched by the corresponding eigenvalues. The area of the unit square changes by $|\lambda_1 \lambda_2|$, the perimeter changes by a factor of $\frac{1}{2}(|\lambda_1| + |\lambda_2|)$.

- SVD: $A = U\Sigma V^T$ — always exists; singular values on diagonal of Σ
- Low-rank approximation: keep top- k singular values (Eckart–Young)
- PCA: eigenvectors of covariance $C = \left(\frac{1}{n}\right)X^T X$
- Connects linear algebra geometry to data compression and denoising

Figure 4.7 SVD and mapping of vectors (represented by discs). The panels follow the same anti-clockwise structure of Figure 4.8.



SVD & PCA Preview — illustration

Vector Calculus & Decompositions — SVD & PCA Preview (source: course materials)

Integration & Differentiation

There and Back Again

- NeurIPS 2020 tutorial: integration **and** differentiation for ML
- Integration: expectations, marginalization, Bayesian evidence
- Differentiation: gradients for optimization, sensitivity analysis
- Map metaphor: numerical integration swamps, Monte Carlo heights, Backprop Bay...

There and Back Again — illustration

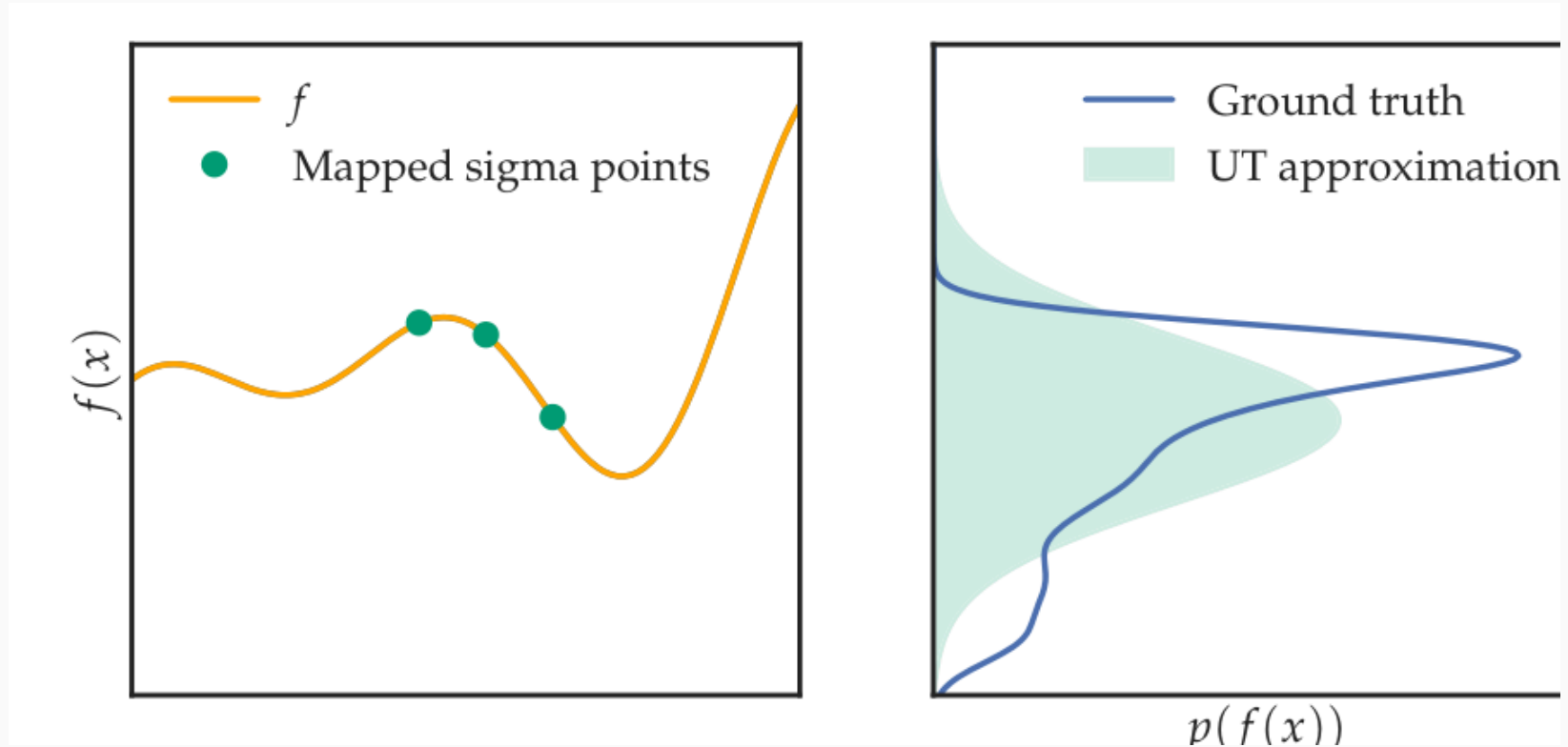


There and Back Again — illustration

Integration & Differentiation — There and Back Again (source: course materials)

- Deterministic quadrature: trapezoid, Simpson, Gauss–Hermite
- Curse of dimensionality motivates Monte Carlo: $\mathbb{E}[f(X)] \approx \left(\frac{1}{N}\right) \sum f(x_i)$
- MC error $O\left(\frac{1}{\sqrt{N}}\right)$ regardless of dimension (with i.i.d. samples)
- Importance sampling & variational methods when f is hard to sample

Integration – Numerical & MC – illustration

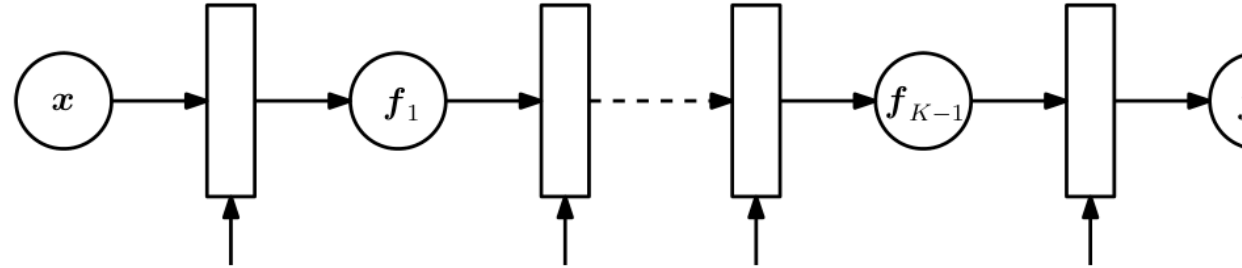


Integration & Differentiation – Integration – Numerical & MC (source: course materials)

- Forward mode: propagate directional derivatives ($O(n)$ for n inputs)
- Reverse mode (backprop): one pass from scalar loss ($O(\text{ops})$ for all params)
- Computational graph stores local Jacobians; chain rule multiplies along paths
- Frameworks implement vector–Jacobian products (VJPs), not full Jacobians

Differentiation — Autodiff & Backprop — illustration

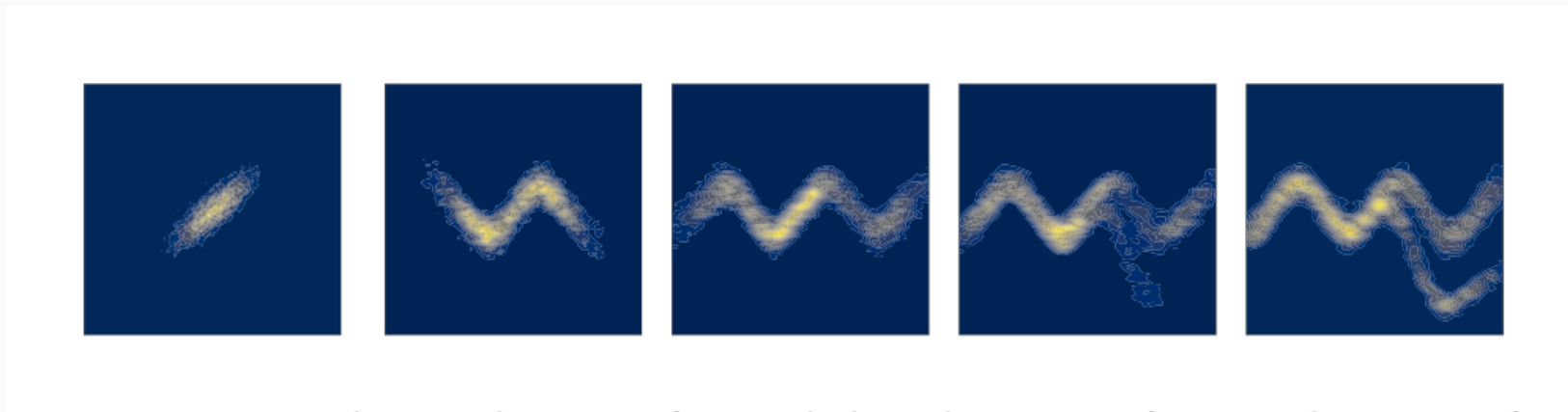
Figure 5.8 Forward pass in a multi-layer neural network to compute the loss L as a function of the inputs x and the parameters A_i , b_i .



Integration & Differentiation — Differentiation — Autodiff & Backprop (source: course materials)

- Implicit Function Theorem: differentiate through $F(x, \theta) = 0$ without explicit solve
- Method of adjoints: efficient gradients for ODE/PDE-constrained objectives
- Lagrange multipliers: constrained optimization via augmented Lagrangian
- Stochastic gradient estimators: REINFORCE, reparameterization trick (Day 6–7)

Implicit Function & Adjoint — illustration

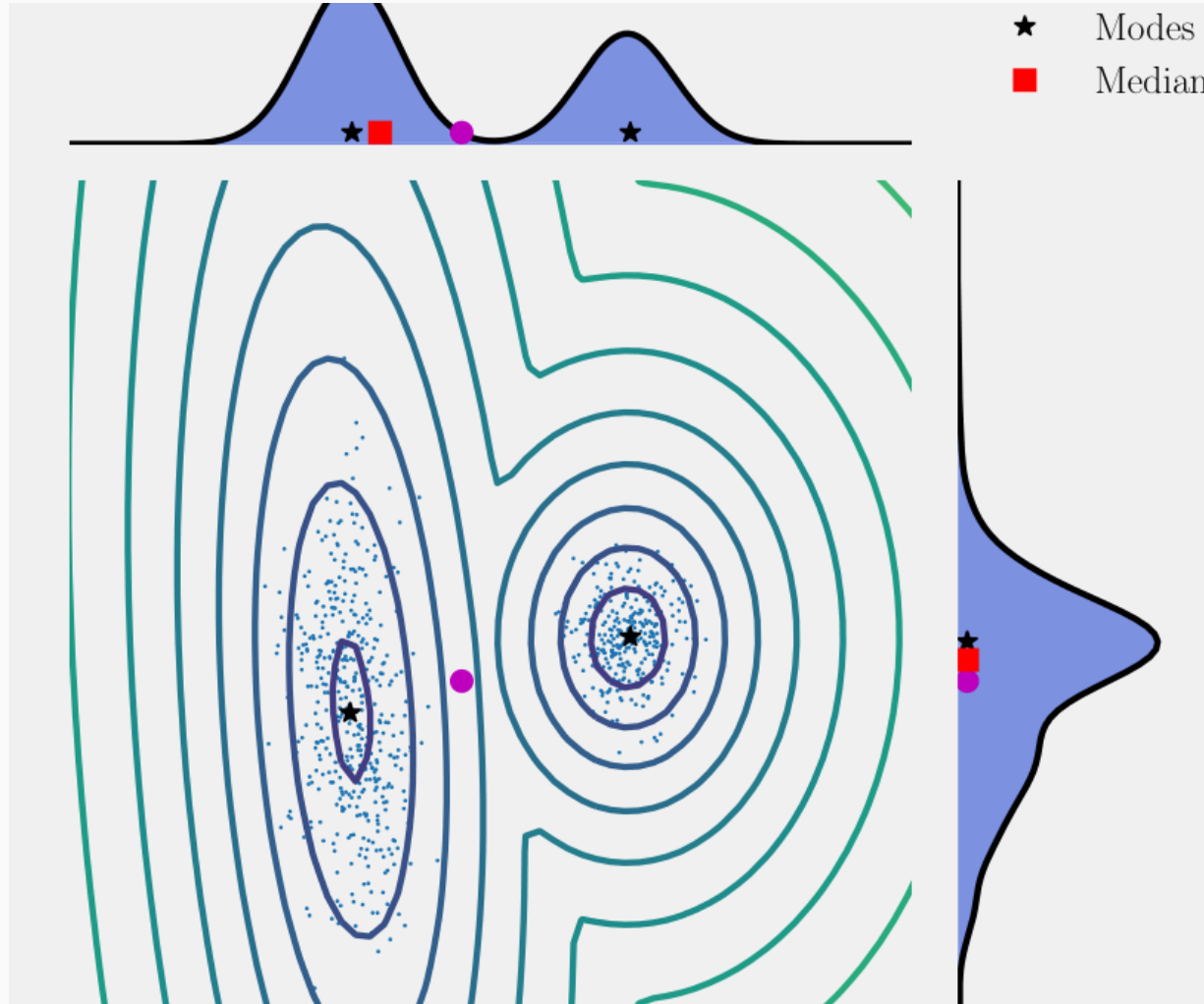


Integration & Differentiation — Implicit Function & Adjoint (source: course materials)

Probability & Distributions

- PMF (discrete) / PDF (continuous); CDF $F(x) = P(X \leq x)$
- Expectation $\mathbb{E}[X]$, variance $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$
- Joint, marginal, conditional; independence $p(x, y) = p(x)p(y)$
- Bayes: $p(\theta|D) \propto p(D|\theta)p(\theta)$

Basics — illustration



median for a two-dimensional dataset, as well as its marginal densities.

Probability & Distributions — Basics (source: course materials)

- $N(x; \mu, \Sigma) = (2\pi)^{-\frac{d}{2}} |\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)\right)$
- Affine transform: if $x \sim N(\mu, \Sigma)$ then $Ax + b \sim N(A\mu + b, A\Sigma A^T)$
- Marginals and conditionals of Gaussians remain Gaussian
- Central role: CLT, linear-Gaussian models, diffusion noise

The Gaussian — illustration

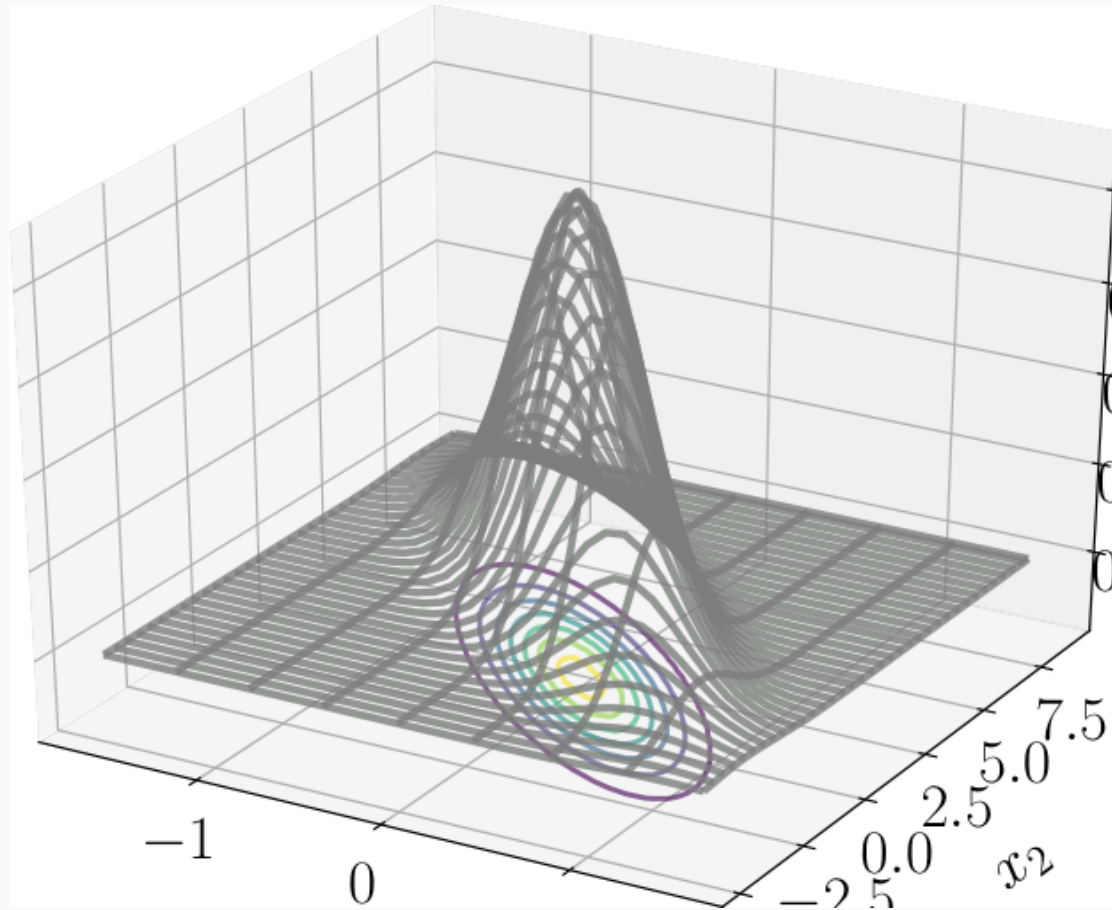


Figure 6.7
Gaussian
distribution of two
random variables x_1
and x_2 .

- Conjugate prior: posterior same family as prior (e.g. Beta–Bernoulli, Normal–Normal)
- Change of variables: $p_{Y(y)} = p_{X(g^{-1}(y))} |\det J_{g^{-1}}(y)|$
- Inverse transform sampling: $x = F^{-1}(u)$ for $u \sim \text{Uniform}(0, 1)$
- Normalizing flows stack invertible maps with tractable Jacobians

Conjugacy & Change of Variables — illustration

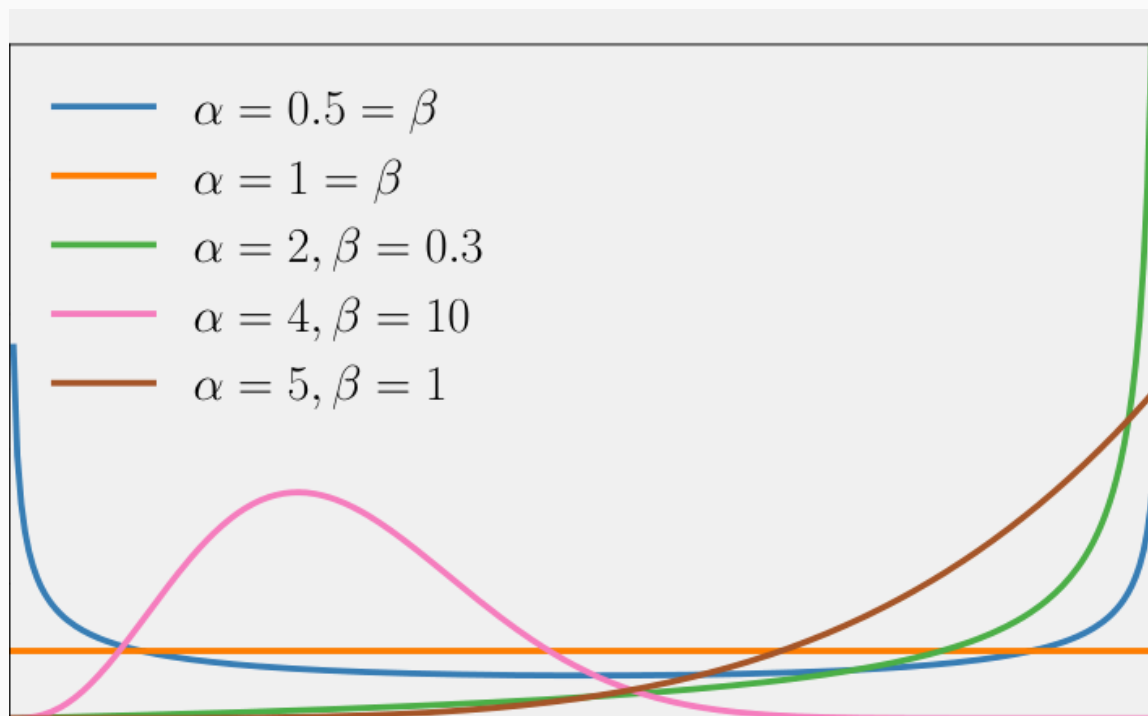


Figure 6.11

Examples of the Beta distribution for different values of α and β .

Probability & Distributions — Conjugacy & Change of Variables (source: course materials)

- Jensen: $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$ for convex f (evidence lower bounds)
- KL divergence: $D_{\text{KL}(q\|p)} = \mathbb{E}_{q[\log \frac{q}{p}]} \geq 0$; not symmetric
- Cross-entropy $H(p, q) = -\mathbb{E}_{p[\log q]}$ — classification loss
- Gaussian KL has closed form — used in VAEs and diffusion training

Differential Equations

- ODE: $\dot{x}(t) = v(x(t), t)$ — rate of change depends on state
- Trajectory view: a curve $x(t)$ through state space
- Vector field view: $v(x, t)$ assigns a velocity at each point
- Same equation, two perspectives — flows vs integral curves

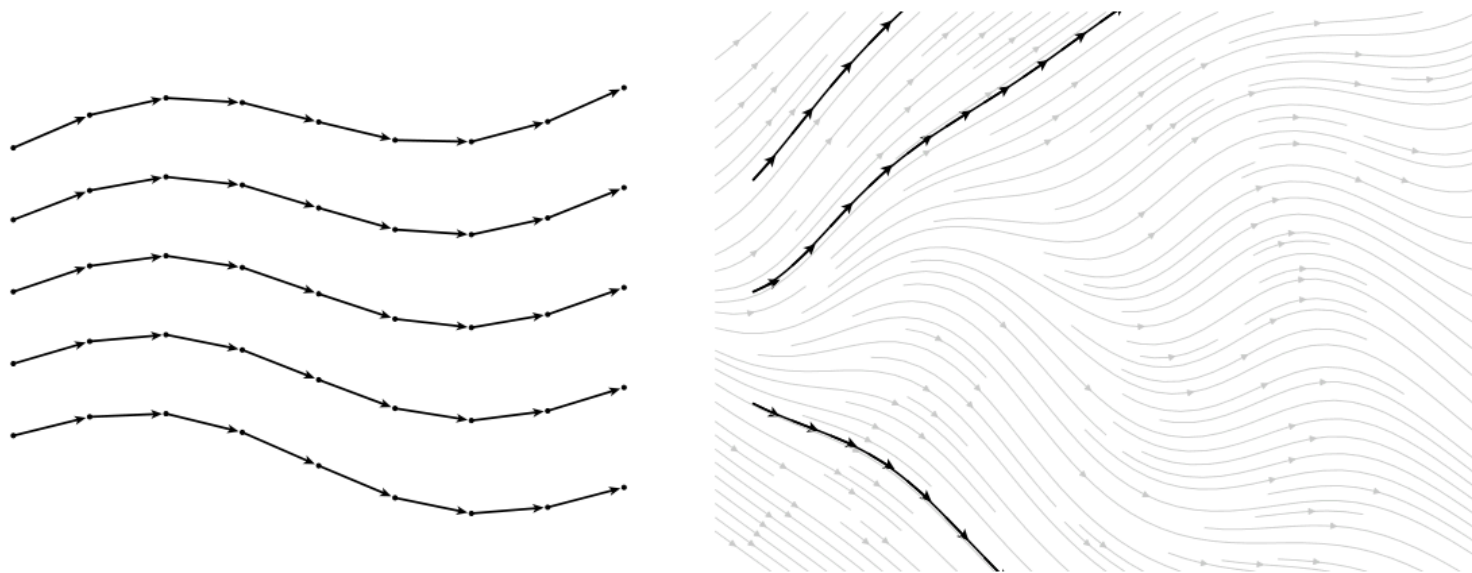


Figure A.1: ODE illustration. A velocity field $\mathbf{v}(\mathbf{x}, t)$ assigns a drift vector at every point. A solution trajectory $\mathbf{x}(t)$ is a path whose tangent always matches the local drift. The left panel shows step-by-step solver updates (dots and arrows) approximating the path, while the right panel shows the exact trajectories (black) flowing consistently with the velocity field. Without specifying an initial state $\mathbf{x}(0)$, there are infinitely many trajectories whose instantaneous changes match the same velocity field. Once $\mathbf{x}(0)$ is fixed, however, the ODE determines a unique path $\mathbf{x}(t)$ that flows according to the drift.

Differential Equations — ODE Intuition (source: course materials)

- Linear ODE: exponential integrator / integrating factor
- Example: $\dot{x} = ax$ gives $x(t) = e^{at}x(0)$
- Existence & uniqueness when v is Lipschitz
- Autonomous vs time-varying; stability from eigenvalues of $\partial \frac{v}{\partial} x$

- Euler: $x_{n+1} = x_n + hv(x_n, t_n)$ — first order, $O(h)$ error
- Heun (RK2): predictor–corrector — $O(h^2)$ local error
- Runge–Kutta (RK4): weighted average of slopes — workhorse $O(h^4)$
- Adaptive step size balances accuracy vs cost (used in neural ODEs, diffusion samplers)

- SDE: $dX_t = f(X_t, t) dt + g(X_t, t) dW_t$ — adds Brownian noise
- Itô calculus: $dW_t^2 = dt$ (quadratic variation)
- Euler–Maruyama: $X_{n+1} = X_n + hf + \sqrt{h}g\xi_n, \xi_n \sim N(0, 1)$
- Foundation for diffusion models (Week 2): forward noising = SDE, reverse = learned drift

- Day 1: **Math Foundations**
- Linear algebra through differential equations — MML Part I
- Questions welcome — practical follows

Questions?

Practical session follows — see course site.